TEST CODE **000592**

FORM TP 9587

JUNE 1995

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

MATHEMATICS

Paper 02 - General Proficiency

2 hours 40 minutes

31 MAY 1995 (a.m.)

INSTRUCTIONS TO CANDIDATES

1. **Answer ALL questions in Section I, and any TWO in Section II.**
2. **Begin the answer for each question on a new page.**
3. **Full marks may not be awarded unless full working or explanation is shown with the answer.**
4. **Mathematical tables, formulae and graph paper are provided.**
5. **Silent electronic calculators may be used for this paper.**
6. **You are advised to use the first 10 minutes of the examination time to read through this paper. Writing may begin during this 10-minute period.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

SECTION I

Answer ALL the questions in this section.

1. **All working must be clearly shown.**

(a) Calculate 0.05181 $\div$ 3.14 and write your answer

(i) correct to 2 decimal places

(ii) correct to 3 significant figures

(iii) in standard form. (4 marks)

(b) Calculate the exact value of

$$(3\tfrac{3}{5} \times 1\tfrac{2}{3}) - 2\tfrac{2}{7}.$$ (3 marks)

(c) The simple interest on a sum of money invested at 3% per annum for 2 years was \$39.75. Calculate the sum of money invested. (3 marks)

Total 10 marks

2. (a) Factorize completely:

(i) $9 - 25m^2$

(ii) $2x^2 - x - 15$

(iii) $x + y - ax - ay$ (6 marks)

(b) Solve $\dfrac{p-1}{2} - \dfrac{p-2}{3} = 1.$ (3 marks)

(c) If $a * b = \sqrt{ab}$, where the positive root is taken, calculate

$(2 * 18) * 24.$ (3 marks)

Total 12 marks

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3. (a) Given that $\underset{\sim}{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\underset{\sim}{b} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$,

(i) sketch vector $\underset{\sim}{a} + \underset{\sim}{b}$.

(ii) express $\underset{\sim}{a} + \underset{\sim}{b}$ as a column vector

(iii) determine the magnitude of $\underset{\sim}{a} + \underset{\sim}{b}$. (4 marks)

(b) A survey of 156 visitors to the Caribbean found that:

118 persons visited Barbados.
98 persons visited Antigua.
110 persons visited Tobago.
25 persons visited Barbados and Antigua only.
35 persons visited Barbados and Tobago only.
30 persons visited Tobago and Antigua only.
x visitors had visited all three countries.

(i) Draw a carefully labelled Venn diagram to represent the information above.

(ii) Write an algebraic expression in x to represent the number of travellers who visited Barbados only.

(iii) Write an equation in x to show the total number of visitors in the survey.

(iv) Calculate the number of travellers who visited all three countries. (7 marks)

Total 11 marks

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4. (a) A straight line is drawn through the points $A(-5, 3)$ and $B(1, 2)$.

(i) Determine the gradient of AB.

(ii) Write the equation of the line AB.

(4 marks)

(b) (i) Using rulers and compasses only, construct

– the triangle CAB with angle $CAB = 60°$, with $AB = 8$ cm and $AC = 9$ cm

– the perpendicular bisector of AB to meet AC at X and AB at Y.

(ii) Measure and state the length of XY.

(iii) Measure and state the size of the angle ABC.

Note: **Credit will be given for construction lines clearly shown.**

(7 marks)

Total 11 marks

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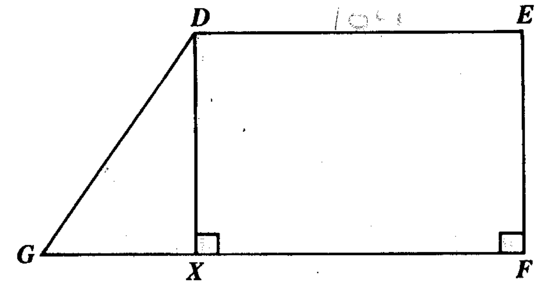
5. (a)



In the trapezium $DEFG$ above, **not drawn to scale**, $DE = 10$ cm, $DG = 13$ cm and $GX = 5$ cm. Angle EFX and angle DXF are right angles.

Calculate

(i) the length of DX

(ii) the area of trapezium $DEFG$. (5 marks)

(b) **In this problem, take π to be $\frac{22}{7}$.**

A piece of wire, formed into a circle, encloses an area of 1386 cm^2.

(i) Calculate the radius of the circle.

(ii) Calculate the length of the wire used to form the circle.

(iii) The wire is then bent to form a square. Calculate, in cm^2, the area of the square. **(7 marks)**

Total 12 marks

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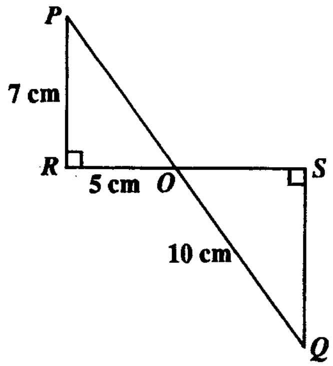
6. (a)



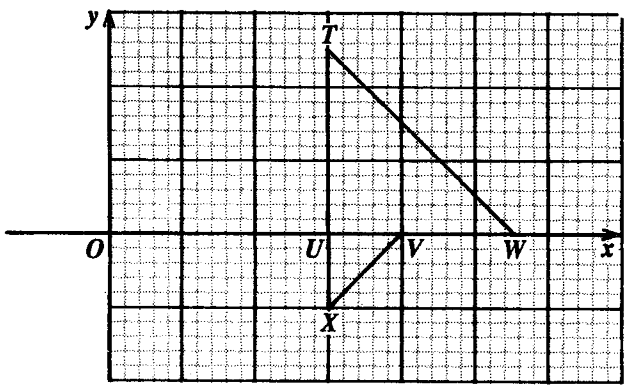
In the diagram above, not drawn to scale, $PR = 7$ cm, $RO = 5$ cm, $QO = 10$ cm and angles at R and S are right angles.

Calculate

(i) the size of angle POR

(ii) the length of RS. (7 marks)

(b)



In the diagram above, the transformation M followed by another transformation N, maps triangle XUV onto triangle TUW.

Describe fully transformations M and N. (5 marks)

Total 12 marks

7. The table below shows the heights of a sample of seedlings measured to the nearest centimetre.

Height (cm)	1 – 3	4 – 6	7 – 9	10 – 12	13 – 15
Frequency	4	14	20	9	3

(a) Calculate

(i) the number of seedlings in the sample.

(ii) the mean height of the seedlings in the sample. (4 marks)

(b) Draw a histogram to display the data. (4 marks)

(c) Calculate the probability that a seedling chosen at random will measure 10 cm or more in height. (3 marks)

Total 11 marks

8. (a) Given $f(x) = \frac{1}{2}x$ and $g(x) = x - 2$,

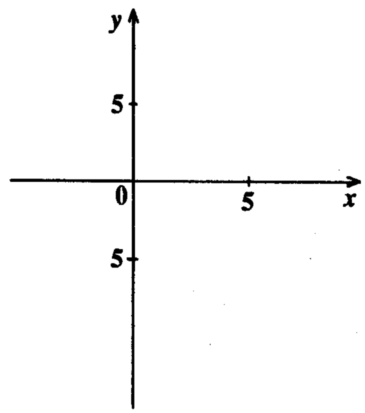
calculate

(i) $g(-2)$

(ii) $fg(4)$

(iii) $f^{-1}(4)$ (6 marks)

(b)



Copy on graph paper the diagram above, and show by shading ONLY, the region which satisfies both the inequalities

$x > 5$

$x \leq y$. (5 marks)

Total 11 marks

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SECTION II

Answer any TWO questions in this section.

RELATIONS AND FUNCTIONS

9. (a) Given π radians $= 180°$, express

(i) $\frac{\pi}{6}$ radians in degrees

(ii) $210°$ in radians. (2 marks)

(b) **Use the answer sheet provided to answer this question.**

A function f is defined on the set of real numbers by

$f(x) = \sin x$.

(i) Complete the table for the given values of x where x is stated in radians and the value of $f(x)$ is given to two places of decimals.

x	0	$\frac{\pi}{6}$	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$	$\frac{5\pi}{6}$	$\frac{6\pi}{6}$	$\frac{7\pi}{6}$
$f(x)$	0	0.50			0.87	0.50		-0.50

(3 marks)

(ii) Using a scale of 2 cm to represent a unit of $\frac{\pi}{6}$ radians on the x-axis and 5 cm to represent 1 unit on the y-axis, draw the graph of $f(x)$ for $0 \leq x \leq \frac{7\pi}{6}$. (6 marks)

(c) By drawing an appropriate straight line on the graph of $f(x)$, estimate the range of values of x (in degrees) for which $\sin x \geq 0.8$. (4 marks)

Total 15 marks

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10. (a) Solve the following equations:

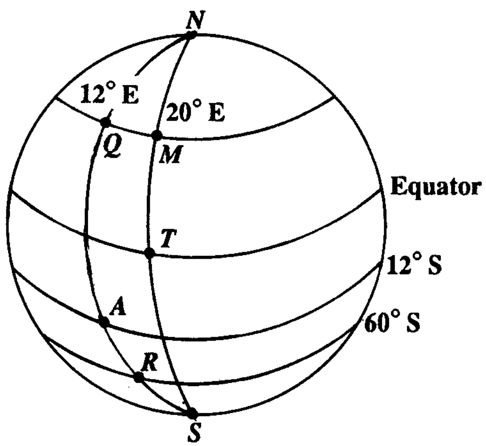
$$x^2 + 9y^2 = 37$$
$$x - 2y = -3.$$
(8 marks)

(b) If $x = 1$ is one root of the equation $(x - c)^2 = 4(x + c + 2)$, calculate to 2 decimal places, the possible values of the constant c. (7 marks)

Total 15 marks

TRIGONOMETRY

11. **In this question, take the radius of the earth to be 6400 km and use $\pi = 3.14$.**



(a) Using the diagram above, **not drawn to scale**, state the coordinates of:

(i) Point A

(ii) Point R

(iii) Point T (3 marks)

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(b) Calculate

(i) the latitude of Q, if Q lies 1600 km due north of A

(ii) the distance in km between Q and M, if M is due east of Q and on longitude 20° E

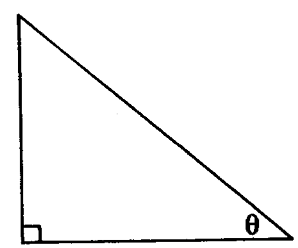
(iii) the longitude of L, if L lies 1200 km due west of R. **(12 marks)**

Total 15 marks

12. (a) Prove the identity

$$\sin^2\theta = (1 + \cos\theta)(1 - \cos\theta)$$
(2 marks)

(b)



In the triangle above, **not drawn to scale**, the right angle and the angle θ are indicated. Given that $\cos\theta = \frac{3}{5}$, calculate the value of $\sin\theta + \tan\theta$.

(5 marks)

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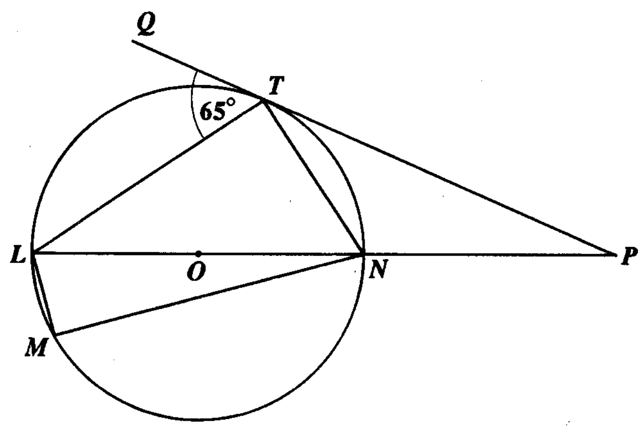
(c)



In the diagram above, **not drawn to scale,** O is the centre of the circle $LMNT$ and PTQ is a tangent to the circle at T. Given that $L\hat{T}Q = 65°$, calculate, stating your reasons, the size of:

(i) $L\hat{M}N$

(ii) $L\hat{N}T$

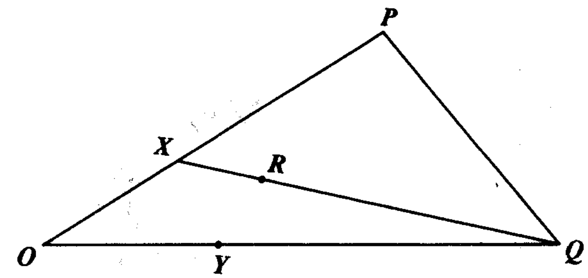
(iii) $T\hat{L}N$

(iv) $T\hat{P}N$ **(8 marks)**

Total 15 marks

VECTORS AND MATRICES

13.



In the diagram above, **not drawn to scale,** $\overline{OP} = 12\boldsymbol{p}$, $\overline{OQ} = 12\boldsymbol{q}$, $\overline{OX} = 4\boldsymbol{p}$ and $\overline{OY} = 4\boldsymbol{q}$.

(a) State $\overline{PQ}$ and $\overline{XY}$ in terms of $\boldsymbol{p}$ and $\boldsymbol{q}$. **(4 marks)**

(b) Given that $\overline{XR} = \frac{1}{4}$ X Q, write $\overline{PR}$ and $\overline{YR}$ in terms of $\boldsymbol{p}$ and $\boldsymbol{q}$. **(8 marks)**

(c) Prove by a vector method that the points Y, R and P are collinear. **(3 marks)**

Total 15 marks

14. (a) If $T = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ and $G = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$,

evaluate TG. (2 marks)

(b) The equation of a straight line PQ is $y = 3x + 1$ and T is a transformation represented by the matrix $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$.

Determine

(i) the value of k, if P is the point $(2, k)$. (3 marks)

(ii) the coordinates of the image of P under the transformation T. (3 marks)

(iii) the coordinates of the point Q if Q is invariant under the transformation T. (7 marks)

Total 15 marks

END OF TEST

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MATHEMATICS

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Answer sheet for Question 9. **Candidate No. ....**

x (radians)	0	$\frac{\pi}{6}$	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$	$\frac{5\pi}{6}$	$\frac{6\pi}{6}$	$\frac{7\pi}{6}$
$f(x)$ (2 dec. places)	0	0.50			0.87	0.50		– 0.50